# R-DKE v2.0 — Physarum-Inspired Recursive Deep Knowledge Engine

*A biologically-inspired architecture for continuous, self-evolving AI knowledge systems*

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## 1. Abstract

Modern AI systems remain primarily reactive — they generate answers only when asked.
R-DKE (Recursive Deep Knowledge Engine) introduced a forward-learning concept in which AI continuously compresses knowledge, constructs a truth graph, interrogates uncertainty, and synthesizes verified answers.

This **v2.0 extension** draws inspiration from *Physarum polycephalum* (slime mold), an organism capable of distributed computation, optimal path-finding, and dynamic resource allocation without centralized control.

We propose a **Physarum-Inspired Reasoning Loop** inside R-DKE, where:

- **Uncertainty = nutrient**
- **Competing reasoning paths grow or decay like slime-mold veins**
- **Verified paths reinforce knowledge graph edges**
- **Incorrect or unproductive paths naturally dissolve**

This enables knowledge to self-organize toward truth-seeking efficiency — forming a living, adaptive cognitive substrate.

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## 2. Background — R-DKE Core Loop (v1)

| Stage | Description |
|---|---|
| Information Intake | World → structured knowledge |
| Semantic Compression | Data → knowledge atoms |
| Truth Graph | Verified, weighted relationships |
| Self-Questioning | Uncertainty triggers recursive inquiry |
| Instant Verified Answers | Answers with full evidence trace |

R-DKE reframes AI from **search** → **think** to **think continuously** → **answer instantly.**

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## 3. Why Physarum?

*Physarum polycephalum* demonstrates:

- Distributed problem solving
- Emergent optimal routing
- Memory without neurons
- Adaptive network reconfiguration
- Efficient exploration/exploitation balance

It has successfully solved:

- Shortest path problems
- Network optimization tasks
- Spanning tree formation

R-DKE v2.0 borrows its **nutrient-driven reinforcement model** to guide reasoning and truth formation.

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## 4. R-DKE v2.0 — The Physarum Loop

**Key Idea**

Knowledge edges behave like **slime-mold veins**:

- **High-confidence edges thicken**
- **Low-evidence edges shrink**
- **Uncertainty flows toward unresolved nodes**
- **Explanations "grow" over time**

In this paradigm, AI does not *choose* answers —
**it grows them.**

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## 5. Algorithmic Sketch

```
for each knowledge_node:
    uncertainty = measure_uncertainty(node)

    if uncertainty > threshold:
        inject("nutrient") into node  // triggers
expansion

    for each connected path:
        propagate_signal(path, strength = truth_weight)

        if path solves contradiction or explains
uncertainty:
            reinforce(path)
```

```
    else:
        decay(path)
```

**Emergent Behavior**

- High-truth pathways strengthen
- Contradictory or ungrounded paths fade
- Hallucinations naturally collapse

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## 6. Expected Capabilities

| Capability | Description |
|---|---|
| Self-stabilizing truth networks | Minimizes hallucination risk |
| Distributed reasoning | No single failure point |
| Adaptive curiosity | System asks: *“Where should I grow next?”* |
| Evidence-driven learning | Strong ideas survive, weak ideas dissolve |
| Long-term emergent intelligence | Memory-like graph evolution |

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## 7. Testing & Validation

Future experiments may simulate:

- Nutrient-flow dynamics on knowledge graphs
- Reinforcement based on truth-confidence scoring
- Path-decay for hallucination-prone reasoning
- Energy-budget-based inference (mirroring slime-mold behavior)

This would allow evaluation of the Physarum Loop’s ability to stabilize truth and suppress false knowledge structures.

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## 8. Conclusion

R-DKE v2.0 proposes a new research direction:
**AI that organizes knowledge like biology — efficient, distributed, and self-improving.**

Instead of querying knowledge, the system **evolves** it.

This architecture may act as a stepping stone toward autonomous reasoning systems beyond transformers and classical search, offering a pathway to more resilient and adaptive intelligence.

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## 9. Citations / Inspirations


- **Nakagaki, T. et al.** — Maze solving by slime mold
- **Tero, A. et al.** — Rules for biologically inspired network design
- **Friston, K.** — Free-energy principle (adaptive systems)
- **Gherasim, M. (2025)** — Early AGI architectures & R-DKE v1